

Day 18: Power BI Visualization Fundamentals

📌 **Definition to Remember**

Data Visualization in Power BI transforms analytical numbers into visual charts to communicate trends, comparisons, proportions, and key performance indicators (KPIs) at a glance.

1. Core Visual Types & When to Use Them

Visual Type	Primary Use Case	Best Practices
Bar / Column Chart	Comparing categorical values (e.g., Sales by Region).	Horizontal Bar for long text labels; Vertical Column for time intervals.
Line Chart	Continuous time-series trends (Monthly Sales YoY).	Keep line count under 4 to avoid visual clutter.
Card / Multi-Row Card	Prominent executive headline metrics (Total Revenue, Orders).	Pair with dynamic YoY delta indicator and clean typography.
Table vs Matrix	Tabular data display.	Matrix supports multi-level row/column hierarchies & subtotals (Pivot Table).
Donut / Pie Chart	Part-to-whole proportions.	Limit to 2-4 slices max. Never use for fine comparisons.
Area Chart	Volume trends over time.	Stacked area shows total magnitude along with component contributions.

2. The New Card Visual (Callout Cards)

- Modern multi-metric card supporting custom subtitle micro-text, accent bars, dynamic SVGs, and sparklines.

★ **Must-Master Skills & Interview Points**

1. Choosing the right visual for the analytical question (Comparison vs Trend vs Distribution vs Composition).
2. Matrix visual features: Drill down, stepped layout, row subtotals, and conditional formatting.
3. Eliminating chart junk: Removing unnecessary gridlines, unformatted decimal places, and duplicate axis titles.

⚡ **Quick Recall**

Bar/Column (Comparison) + Line (Trend over time) + Matrix (Hierarchies) + Cards (Headline KPIs)